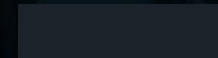
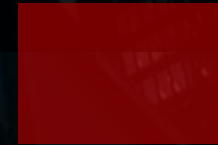




The Shift to AI-Driven BI & Snowflake Intelligence





AGENDA

Evolution of BI

BI Maturity Model

Why Conversational BI?

Snowflake Intelligence

NLQ Fundamentals

Prompt Engineering for UI

Architecture Overview



The Evolution of Business Intelligence

1

1990s - Reporting Era

Static reports, scheduled queries, IT-dependent

2

2000s - OLAP Era

Pivot tables, cubes & MDX, slice & dice

3

2010s - Self-Service Era

Drag-and-drop, Tableau, Power BI, business-led

4

2020s - AI-BI Era ← We are here

Conversational, NL queries, agent-driven

Static Dashboards vs. Conversational Analytics

Static Dashboard

- Fixed KPI tiles that never change
- Users must navigate to find answers
- IT bottleneck for new metrics
- No context-awareness per user role
- Data shown, not explained
- Refresh cycles create stale data

Conversational AI-BI

- Ask questions in plain English
- AI surfaces the right answer instantly
- Self-service for any business user
- Persona-aware: exec or analyst views
- Explains trends with business context
- Real-time Snowflake queries on demand

BI Maturity Model - Where Does Your Team Stand?

1

L1 - Descriptive

What happened? Historical reports, standard KPIs

2

L2 - Diagnostic

Why did it happen? Drilldowns, root cause analysis

3

L3 - Predictive

What will happen? ML models, forecasting

4

L4 - Prescriptive

What should I do? AI recommendations, automated actions

5

L5 - Conversational

Talk to your data - NLQ + agent-driven AI insights

Why Conversational BI? The Business Case

73%

**Dashboard
Abandonment**

Business users quit within 2 clicks

5x

Faster Decisions

NLQ vs. traditional BI navigation

60%

IT Ticket Reduction

When users self-serve with AI

\$2.4B

Market Size 2025

Conversational analytics growing 24% YoY



What is Snowflake Intelligence?

Snowflake's native AI layer - business users interact with their data warehouse using natural language. No SQL required.

Conversational Interface

Chat-like UI - type questions, get data-backed answers

Cortex AI Engine

LLM-powered query generation: COMPLETE, SUMMARIZE, CLASSIFY

Auto-Visualization

Renders charts, tables, or summaries based on query context

Snowflake Cortex: Core AI Functions

SNOWFLAKE.CORTEX.COMPLETE()

Calls an LLM with a prompt. Used for summaries, explanations, and classifications.

```
SELECT SNOWFLAKE.CORTEX.COMPLETE('llama3-70b',  
  'Summarize this sales data: ' || sales_text);
```

SNOWFLAKE.CORTEX.SUMMARIZE()

Condenses long text fields. Great for support tickets and call logs.

```
SELECT SNOWFLAKE.CORTEX.SUMMARIZE(transcript_column)  
FROM support_tickets;
```

SNOWFLAKE.CORTEX.CLASSIFY_TEXT()

Classifies text into user-defined categories. Ideal for sentiment and intent analysis.

```
SELECT SNOWFLAKE.CORTEX.CLASSIFY_TEXT(feedback,  
  ['Positive', 'Negative', 'Neutral']);
```

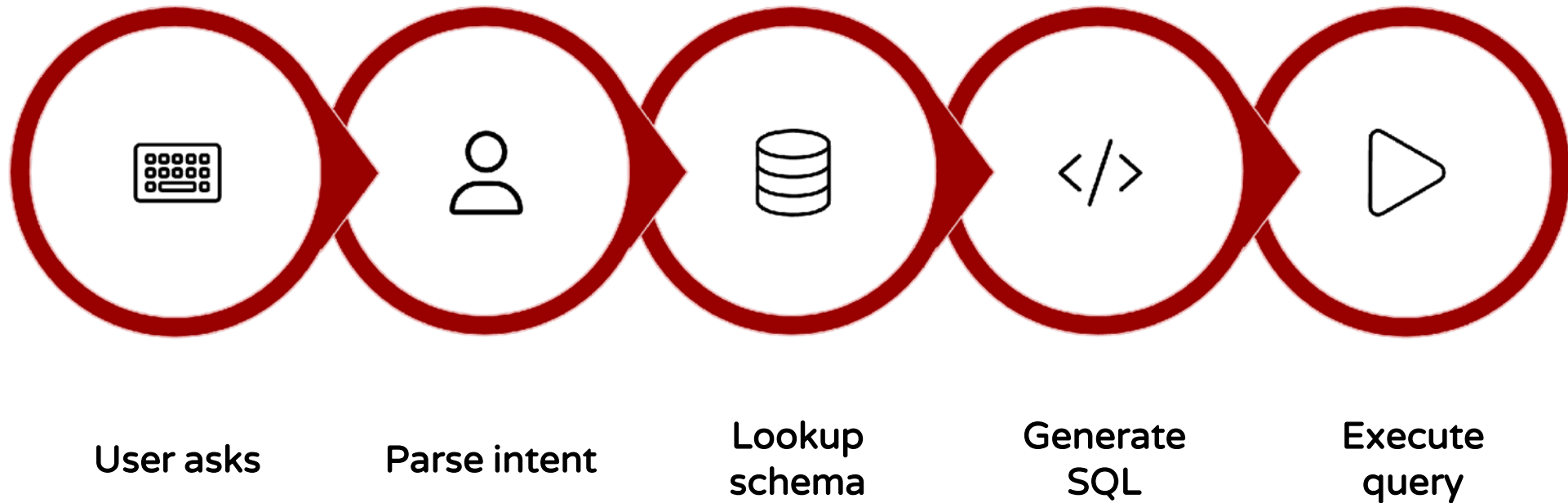
SNOWFLAKE.CORTEX.TRANSLATE()

Translates text between languages in SQL. Useful for global event data.

```
SELECT SNOWFLAKE.CORTEX.TRANSLATE(comments, 'de', 'en')  
FROM event_feedback;
```


Natural Language Query (NLQ) - Fundamentals

NLQ bridges human language and database query language - users type or speak, the system generates SQL.



Example: *"Show me the top 10 events by registration count in Q1 2025 where attendance dropped vs Q1 2024"* → The NLQ engine resolves entities, joins tables, applies filters, and returns a ranked result set - all without a single line of SQL written by the user.

NLQ Engine - Key Components



Intent Parser

Identifies query type: aggregation, ranking, time comparison, trend, or lookup. Uses few-shot prompting to map user language to query patterns.



Schema Resolver

Looks up the data catalog or semantic layer to find exact tables, joins, and relationships needed to answer accurately.



Result Formatter

Decides how to present results: chart, data table, or natural language summary - based on query type and persona.



Entity Extractor

Picks out key nouns - table names, column names, date ranges. Maps business terms like "revenue" to actual DB columns like TOTAL_REVENUE_USD.



SQL Generator

Uses LLM (Llama 3 via Cortex) to generate syntactically correct Snowflake SQL, including CTEs, window functions, and group-by clauses.



Feedback Loop

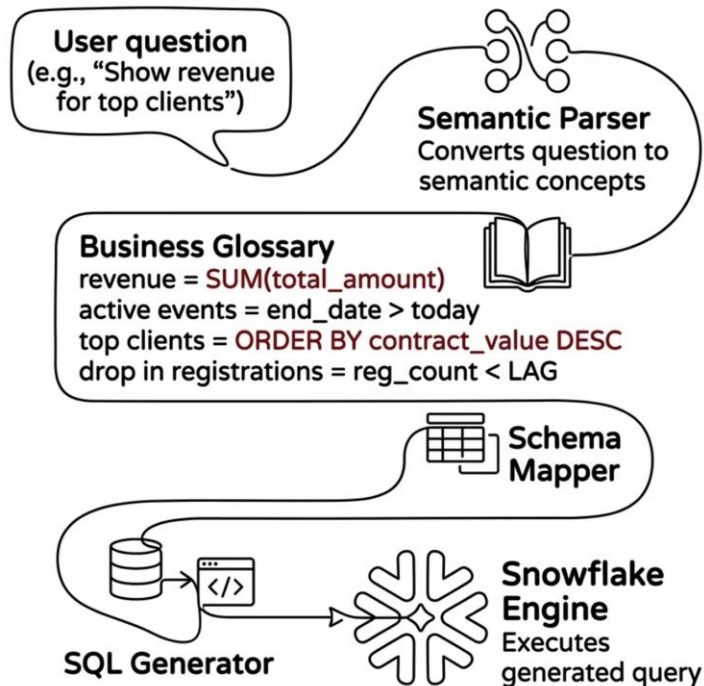
Captures user corrections and thumbs up/down to re-train the semantic layer mappings and improve accuracy over time.

The Semantic Layer - NLQ's Secret Weapon

What is a Semantic Layer?

A semantic layer maps business language to technical database objects - it's the bridge that lets an LLM understand that "revenue" means `SUM(order_amount)` from the SALES table.

- Without a semantic layer, the LLM guesses at schema. With one, it knows exactly which tables, joins, and filters apply.



Prompt Engineering for BI Interfaces

The LLM response format must be engineered to fit the UI - charts need JSON, summaries need prose, and tables need structured data.

1

Chart Response

Return ONLY a JSON object with chart_type, title, x_axis, y_axis, data. No prose.

→ *Rendered as bar/line/pie chart in React*

2

Executive Summary

Exactly 3 bullets, each starting with a metric. No charts.

→ *Rendered as styled summary card for executives*

3

Analyst Drilldown

Markdown table: Metric | Value | Change | Trend. Include statistical significance.

→ *Rendered as sortable data table*

4

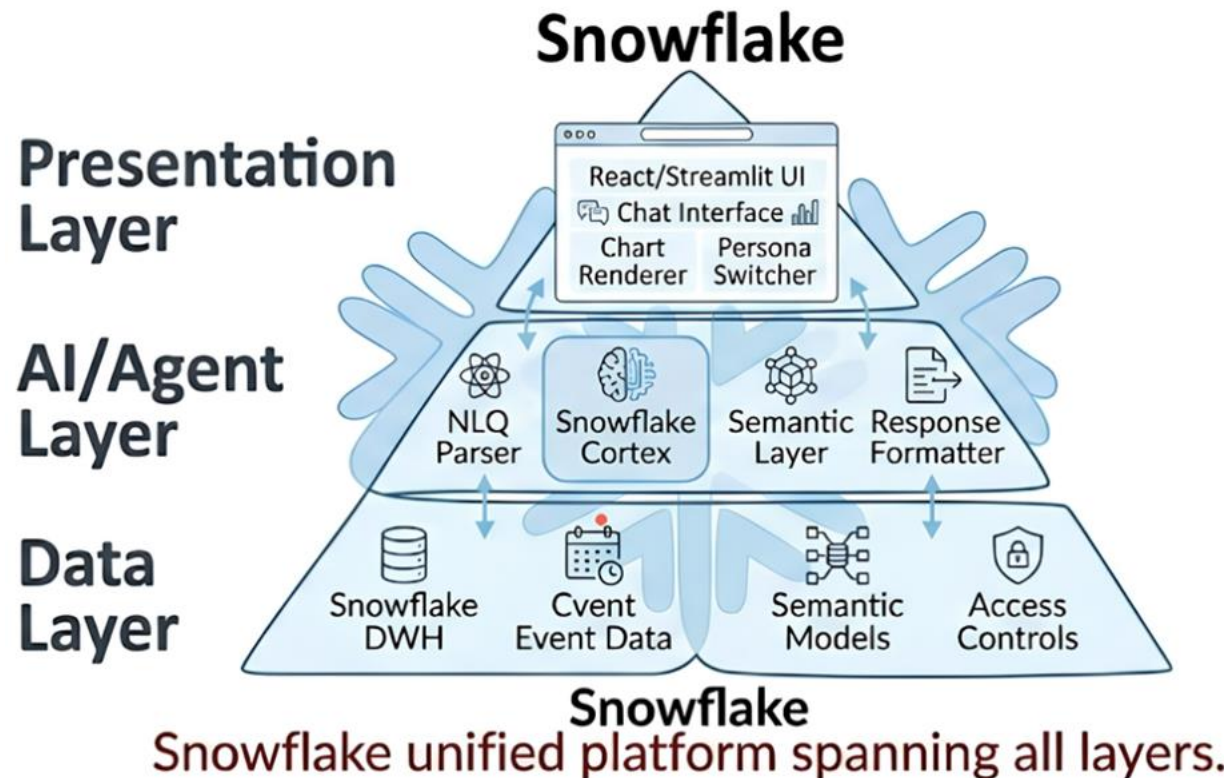
Action Suggestion

After data, append "Recommended Actions:" with 2–3 specific, actionable next steps.

→ *Rendered as action cards with CTA buttons*

Snowflake Cortex - Available LLM Models

Snowflake Cortex provides access to multiple LLMs via a unified API - choose the right model for cost, speed, and capability.



Snowflake provides the unified platform spanning all three layers - data warehouse + AI engine + application runtime.

Snowflake Intelligence vs. Traditional BI Tools

Feature	Tableau / Power BI	Snowflake Intelligence
Query Interface	Point & click / drag-drop	Natural language chat
SQL Required?	Yes (for custom queries)	No - AI generates SQL
Persona Adaptation	Manual filter panels	AI adapts to user role
Real-time Data	Scheduled refreshes	Live Snowflake queries
Custom AI Logic	Limited (external tools)	Native Cortex functions
Deployment	SaaS / on-prem server	Runs inside Snowflake
Extensibility	Embed with tokens	Streamlit / React / API

Snowflake Cortex - Available LLM Models

Snowflake Cortex provides access to multiple LLMs via a unified API - choose the right model for cost, speed, and capability.



Llama 3.1 405B

Complex reasoning, multi-step NLQ, code generation

Speed: Medium | Cost: High



Llama 3.1 70B

General BI queries, summarization, classification

Speed: Fast | Cost: Medium



Llama 3.1 8B

Simple lookups, quick categorization, high-volume tasks

Speed: Very Fast | Cost: Low

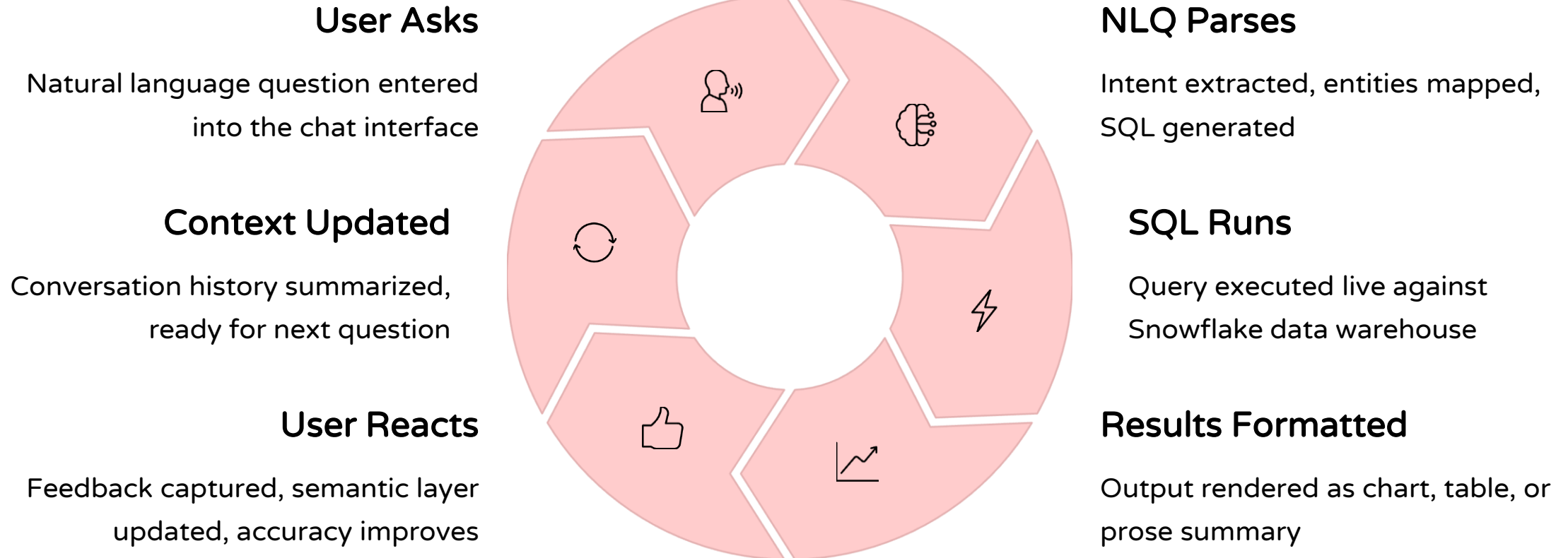


Mistral Large

Instruction-following, chart prompt engineering

Speed: Fast | Cost: Medium

The Conversational BI Loop



Real-World Use Case: Event Analytics

An event planner wants to understand registration trends - no SQL, no complex dashboards.

Q: How many events ran last quarter?

A: 147 events in Q4 2024, avg. attendance 312 - **18% higher than Q4 2023**

Q: Which event type had the most cancellations?

A: Virtual: 23% | Hybrid: 17% | In-person: 8% cancellation rate

Q: Show monthly registration trends for my top 5 events

A: Multi-line chart rendered - Jan through Dec, 5 trend lines with legend



Snowflake Intelligence - UI Components

Data Sources Connected

- CVENT_EVENTS
- REGISTRATIONS
- VENUES
- REVENUE_DATA

Ask a question...

"Show top events by revenue in 2024"



AI Response

Here are your top 5 events by revenue in 2024:

1. Dreamforce Summit - \$4.2M
2. TechConnect Annual - \$3.8M
3. HR Leaders Forum - \$2.9M



Results pulled live from Snowflake • Refreshed 2h ago • Source: REVENUE_DATA table

How AI-BI Changes Every Role

Business Analyst

Before: Tickets to IT, days of waiting, pre-built dashboards only

After: Self-serves with NLQ, instant answers, personalized views

Data Scientist

Before: Runs analysis in notebooks, hand-delivers findings

After: Publishes models as AI skills, automates insight delivery at scale

BI Engineer

Before: Builds static dashboards, wrangles ad-hoc requests

After: Designs AI agent systems, builds semantic layers, focuses on architecture

Executive

Before: Reads PDF reports weekly, misses real-time opportunities

After: Asks questions directly in chat, acts on AI-suggested actions

Key Design Principles for AI-BI Systems



01 - Trust by Default

Always show data lineage: "This answer came from EVENTS table, Q1 2024, refreshed 2h ago." Users must trust the source before they act.



02 - Fail Gracefully

When NLQ can't parse a query, suggest alternatives: "Did you mean [A] or [B]?" Never surface raw errors to the user.



03 - Context is King

Every query carries conversation context. "Show me last month's data" after discussing events must resolve to event registrations.



04 - Progressive Disclosure

Start with the headline. Let users drill down on demand. Don't overwhelm with raw data.



05 - Persona Awareness

The same query returns different formats: an analyst gets a filterable table; an exec gets 3 bullets and a chart.



06 - Actionability

"Registration is down 20% - consider sending a reminder email to [X segment]." The best AI-BI systems suggest what to do next.

Snowflake Intelligence - Event-Specific Use Cases

Event Performance Analytics

- Which events had the highest ROI last quarter?
- Attendance trends: virtual vs. in-person?

✓ Instant benchmark visibility without SQL

Attendee Behavior Analysis

- Average session engagement score by event type?
- Which segments have highest repeat registration rate?

✓ Persona-level insights on demand

Revenue Intelligence

- Which pricing tiers drive the most total revenue?
- Month-over-month revenue growth for enterprise clients?

✓ Real-time financial dashboards in chat

Operational Efficiency

- Which events required the most support tickets?
- Average setup time vs. event size - any correlation?

✓ Operations team self-serves without IT

NLQ SQL Quality: Good vs. Bad Prompt Engineering

✖ Poor Prompt

System: You are a SQL assistant.
Convert user questions to SQL.

User: show me top events

Result: `SELECT * FROM events LIMIT 10;`

⊗ Missing: date filter, metric definition, sort order, business context

✔ Strong Prompt

System: Snowflake SQL expert for Cvent.
Tables: EVENTS(id, name, type, start_date),
REGISTRATIONS(event_id, count, date).
Always filter by current year. Return
top-N by registration_count DESC.

User: show me top events

Result: `SELECT e.name,
SUM(r.count) AS registrations
FROM EVENTS e JOIN REGISTRATIONS r
ON e.id=r.event_id
WHERE YEAR(e.start_date)=2025
GROUP BY e.name
ORDER BY registrations DESC LIMIT 10;`

Understanding Tokens & Context Windows in AI-BI

Every LLM call has a token budget. Smart AI-BI systems manage what goes into the prompt vs. what gets retrieved at runtime.

What Are Tokens?

Chunks of text (~4 chars each). "registration count" = 4 tokens. Every character in your system prompt, schema, history, and query counts against your budget.

Context Window Limits

Llama 3.1 70B: 128K tokens. A large schema dump + long conversation history can easily hit this limit, causing truncation errors.

Schema Injection Strategy

Don't inject the full DB schema. Use the semantic layer to inject only RELEVANT tables and columns based on the user's question intent.

Conversation Summarization

For multi-turn chats, summarize older turns into compact state: "User has been asking about Q1 2025 event performance" - rather than keeping full history.

AI-BI Performance: Latency Optimization Strategies

Optimization Checklist

- ✓ Cache schema metadata - refresh every 24h, not per query
- ✓ Pre-warm Cortex with common query patterns during off-peak hours
- ✓ Use streaming responses - show partial results as tokens arrive
- ✓ Route simple queries to 8B model, complex ones to 70B
- ✓ Use Snowflake result cache for identical query reuse

AI-BI Security: Role-Based Access in Snowflake

❌ Security must be enforced at the **Snowflake layer**, NOT the application layer. The AI agent runs under the user's Snowflake role - it can only query what that role can access.

🛡️ EXECUTIVE_ROLE

Read: summary views, aggregated KPIs

No PII, no individual records, no raw tables

🛡️ ANALYST_ROLE

Read: all business tables, drilldown data

Masked PII - emails truncated

🛡️ ENGINEER_ROLE

Read/Write: all tables, raw data

Full access including PII columns

🛡️ READONLY_ROLE

Read: published dashboards only

Pre-approved semantic views

Snowflake Intelligence - Integration Ecosystem



Frontend

- Streamlit in Snowflake
- React / Next.js
- Node.js API layer
- VS Code Snowflake Ext



Data Sources

- Snowflake DWH
- Snowflake Iceberg
- External Stages (S3)
- Kafka Streaming



BI Tools

- Sigma Computing
- Tableau CRM
- Power BI Connector
- Looker Studio



AI / ML

- Snowflake Cortex
- MLflow on SPCS
- dbt Semantic Layer
- TruLens Evals

AI-BI Response Format Engineering

Chart JSON

Triggered when: user asks "show me", "chart", "visualize", "trend", "compare"

```
{"type": "bar", "title": "Top Events",  
  "labels": ["A", "B", "C"], "values": [120, 98, 87]}
```

→ React chart component

Prose Summary

Triggered when: user asks "explain", "why", "summarize", "what does this mean"

Registration dropped 22% in October due to 3 conflicting industry events. Recovery projected by December.

→ Styled text card component

Data Table

Triggered when: user asks "list", "show all", "give me details", "breakdown"

```
{"columns": ["Event", "Date", "Registrations"],  
  "rows": [{"Summit", ...}]}
```

→ Sortable HTML table component

Action Card

Triggered when: anomaly detected or user asks "what should I do", "recommend"

Action: Send reminder email | Target: 245 pending registrants |
Expected lift: +18%

→ CTA card with action button

Evaluating NLQ Quality: Metrics That Matter

Metric	Definition	Target	How to Measure
SQL Execution Rate	% of NLQ queries producing valid SQL	> 95%	Log failed SQL executions
Answer Correctness	% matching human-verified ground truth	> 85%	TruLens RAG evaluation
Context Relevance	SQL reflects the question's actual intent	> 90%	LLM-as-judge scoring
Latency P95	95th percentile end-to-end response time	< 3s	Snowflake query profiler
User Satisfaction	Thumbs up rate on AI responses	> 75%	In-app feedback widget
Hallucination Rate	% citing data not in actual query results	< 2%	Output vs. source comparison

Snowflake Cortex Analyst: NLQ Built-In

Cortex Analyst is a fully managed NLQ service inside Snowflake - generates SQL from natural language using a semantic model YAML file. No SDK required.

1

Define Semantic Model

Create a YAML file describing tables, metrics, dimensions, and business terms - your data dictionary for the AI.

2

Upload to Snowflake Stage

PUT the YAML into a Snowflake internal or external stage. Cortex Analyst reads from this stage at runtime.

3

Call REST API

POST to `/api/v2/cortex/analyst/message` with messages and `semantic_model_file`. No SDK required.

4

Receive SQL + Answer

Response includes generated SQL, natural language answer, and chart data - ready to render in your UI.

Data Freshness & Caching Strategies

Live Queries

Real-time data critical (live event dashboards)

Latency: 2–10s | Freshness: 100% | Cost: High

✓ Always current ✗ Slow at scale



Result Cache

Same query likely repeated (executive summaries)

Latency: <100ms | Freshness: Up to 24h | Cost: Low

✓ Near-instant ✗ Stale if tables change

Materialized Views

Complex aggregations queried frequently

Latency: <500ms | Freshness: Scheduled | Cost: Medium

✓ Pre-computed, fast joins ✗ Maintenance overhead



Semantic Cache

Same NLQ question asked multiple times

Latency: <50ms | Freshness: Configurable | Cost: Very Low

✓ Skip LLM call entirely ✗ Requires similarity matching



A low-angle, upward-looking photograph of several modern skyscrapers with glass facades. The image is overlaid with a semi-transparent dark grey horizontal band across the middle. Three solid red rectangular blocks are positioned: one at the top center, one at the bottom left, and one at the bottom right. The text 'THANK YOU' is centered within the dark band in a white, bold, sans-serif font.

THANK YOU